Nerusupalli Rampanikar Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.21
Sri Chaitanya Junior College	Class XII (BIEAP)	2023	90.00%
Sri Chaitanya High School	Class X (SSC)	2021	10.00%

SCHOLASTIC ACHIEVEMENTS

Qualified for **PRMO** (Pre-Regional Mathematical Olympiad), conducted by HBCSE, among top mathematics students selected for the national Olympiad track

SKILLS

Programming Languages: C/C++, Python

Web Technologies: HTML, CSS, Node.js

Databases & Storage: MySQL

Tools: Git, GitLab, Linux, Bash, LaTeX, Make, GDB, Solid Edge, awk

RELEVANT COURSES

CSE Core: Data Structures, Operating Systems, Database Management, Computer Networks, Compilers, Foundations of Machine Learning, Deep Learning, Fraud Analytics, Multi-Agent Systems

Mathematics & EE: Discrete Mathematics, Calculus, Probability and Random Variables, Digital Circuits

PROJECTS

CareFlow: AI Care Coordination Assistant

AI / Machine Learning Project

- Engineered **LangGraph-based multi-agent workflows** using state-machine orchestration to coordinate specialized agents across care-coordination tasks.
- Built a **Retrieval-Augmented Generation (RAG) pipeline** grounded in clinical guidelines with guardrails for unsafe diagnostic requests.
- Integrated external tools through **MCP-based tool calling** and implemented end-to-end tracing for accuracy, latency, task completion, and reliability.

Role Management System (RBAC)

Software Engineering Project | Group Project

- Developed a centralized authorization platform using **Node.js, Express, MongoDB, React, and Module Federation** for distributed microservices and microfrontends.
- Implemented **RBAC/ABAC policy evaluation** with **JWT authentication and Redis-based permission caching**.
- Added temporary role assignments, API endpoint registration, microfrontend access control, and audit logging with permission revocation and expiry handling.

Adaptive Latency-Aware Handshaking Protocol (ALHP)

Course Project | Prof. Kotaro & Prof. Praveen | Aug 2025 - Nov 2025

- Designed an **adaptive transport protocol for short-lived IoT connections** that selects handshake modes using

RTT, packet-loss, and confidence metrics.
- Implemented **state machines, RTT/loss estimation, session tokens, retransmission, escalation, and hysteresis-based mode selection**.
- Built an **NS-3 and real-socket evaluation framework** to analyze varying network conditions and compare performance against TCP and UDP.

Multi-Client Quiz System with LLM Integration

Course Project | *Prof. Kotaro & Prof. Praveen* | *Aug 2025 - Nov 2025*

- Built a **C++ TCP-based multi-client quiz system** with LLM integration for automated question generation, answer evaluation, and leaderboard management.
- Designed a **custom application-layer protocol** with structured messaging and keep-alive mechanisms for reliable client-server communication.
- Implemented **FIFO server-side caching** and evaluated latency and throughput under concurrent clients using Wireshark.

RISC-V Assembly Simulator with Cache

Course Project | *Prof. Rajesh* | *Aug 2024 - Nov 2024*

- Developed a **C++ RV64I instruction-set simulator** supporting sequential execution, register-state tracking, control flow, and breakpoint-based debugging.
- Implemented a configurable **cache simulator** supporting varying cache sizes and associativities to model memory-access behavior.
- Added **LRU, LFU, and Random replacement policies** with cache statistics to analyze cache performance across configurations

EXTRACURRICULAR

Sports: Volleyball

Social Service: NSS Blood Donor